Version No.			

ROLL NUMBER						

Answer Sheet No._____

Sign. of Candidate _____

Sign. of Invigilator _____

MATHEMATICS HSSC–II (3rd Set)

SECTION – A (Marks 20)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. **Do not use lead pencil.**

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

(1) If $f(x) = \frac{2x+1}{2x-1}$, then $f^{-1}(x)$ is:

A. $\frac{1}{2}\left(\frac{x+1}{x-1}\right)$ ◯ B. $\frac{1}{2}\left(\frac{x-1}{x+1}\right)$ ◯

C. $\frac{1}{2}\left(\frac{x+2}{x-2}\right)$ ◯ D. $\frac{1}{2}\left(\frac{2x-1}{2x+1}\right)$ ◯

(2) The value of $\lim_{x\to a} \frac{x^7-a^7}{x-a}$ is:

A. 0 ◯ B. $\frac{0}{0}$ ◯

C. $7a^7$ ◯ D. $7a^6$ ◯

(3) The value of $\lim_{x\to 0} \frac{\sin(sinx)}{x}$ is:

A. 0 ◯ B. 1 ◯

C. ∞ ◯ D. e ◯

(4) If $f'(x) = \frac{x}{1+x}$ $\forall\, x \geq 0$, then $f(x)$ is said to be:

A. stationary ◯ B. increasing ◯

C. decreasing ◯ D. constant ◯

(5) If $y = 7^{x^3}$, then $\frac{dy}{dx}$ is:

A. $(x^3)(7)^{x^3-1}$ ◯ B. $7^{x^3}(3x^2)$ ◯

C. $x^3 ln7$ ◯ D. $7^{x^3}(3x^2 ln7)$ ◯

(6) The derivative of $cosx$ w.r.t. $tanx$ is:

A. $sinx \cos^2 x$ ◯ B. $-sinx\ cosx$ ◯

C. $-sinx \cos^2 x$ ◯ D. $sinx\ sex^2 x$ ◯

(7) The value of $\int_1^e \frac{lnx}{x} dx$ is:

A. $2\ln(e)$ ◯ B. $\ln(e)$ ◯

C. $\frac{1}{2}$ ◯ D. $-\frac{1}{2}$ ◯

(8) The anti-derivatives of $(x-1)e^{-x}$ is:

A. $-xe^{-x}+c$ ◯ B. $xe^{-x}+c$ ◯

C. $xe^{x}+c$ ◯ D. $-xe^{-x}+e^{-x}+c$ ◯

(9) The integral of $\frac{sin2x}{sinx}$ is:

A. $2sinx$ ◯ B. $-2sinx$ ◯

C. $\frac{1}{2}sinx$ ◯ D. $-\frac{1}{2}sinx$ ◯

(10) The perpendicular distance from origin to the line $xcos30^o + ysin30^o = 1$ is:

A. 2 ◯ B. 0.5 ◯

C. 0.866 ◯ D. 1 ◯

(11) The distance between $6x+8y-15=0$ and $3x+4y-9=0$ is:

A. $3/2$ ◯ B. $3/4$ ◯

C. $3/10$ ◯ D. $2/7$ ◯

(12) The slope of the line bisecting second and fourth quadrant is:

A. 1 ◯ B. -1 ◯

C. 0 ◯ D. ∞ ◯

(13) The point where two boundary lines of a shaded region intersects is called:

A. boundary point ◯ B. corner point ◯

C. stationary point ◯ D. feasible point ◯

(14) Linear inequality $2x-3y>5$ is satisfied by the point:

A. (2,1) ◯ B. $(-2,3)$ ◯

C. $(-3,-5)$ ◯ D. (3,2) ◯

(15) The circle $x^2+y^2-8x+4y+4=0$ passes through:

A. I and II quadrants ◯ B. II and III quadrants ◯

C. III and IV quadrants ◯ D. I and IV quadrants ◯

(16) The equation of ellipse with foci $(\pm 2,0)$ and eccentricity $\frac{1}{2}$ is:

A. $\frac{x^2}{12}+\frac{y^2}{16}=1$ ◯ B. $\frac{x^2}{16}+\frac{y^2}{12}=1$ ◯

C. $\frac{x^2}{16}+\frac{y^2}{8}=1$ ◯ D. $\frac{x^2}{8}+\frac{y^2}{16}=1$ ◯

(17) The parametric equations of the parabola are:

A. $x=acos\theta\,, y=asin\theta$ ◯ B. $x=asin\theta\,, y=bcos\theta$ ◯

C. $x=a\theta^2\,, y=2a\theta$ ◯ D. $x=asec\theta\,, y=btan\theta$ ◯

(18) The direction cosines of the vector $3\hat{\imath}-\hat{\jmath}-4\hat{k}$ is:

A. $\left(-\frac{3}{\sqrt{26}},\frac{1}{\sqrt{26}},\frac{4}{\sqrt{26}}\right)$ ◯ B. $\left(\frac{3}{\sqrt{26}},-\frac{1}{\sqrt{26}},-\frac{4}{\sqrt{26}}\right)$ ◯

C. $\left(-\frac{3}{\sqrt{26}},\frac{1}{\sqrt{26}},-\frac{4}{\sqrt{26}}\right)$ ◯ D. $\left(-\frac{3}{\sqrt{26}},-\frac{1}{\sqrt{26}},\frac{4}{\sqrt{26}}\right)$ ◯

(19) If the volume of a parallelopiped with $\underline{a}$, $\underline{b}$ and $\underline{c}$ as co-terminus edges is 96, then volume of the tetrahedron with same edges is:

A. 48 ◯ B. 32 ◯

C. 24 ◯ D. 16 ◯

(20) If $\underline{a}$ and $\underline{b}$ be any two vectors along the adjacent sides of the triangle, then the area of triangle is:

A. $\underline{a}\times\underline{b}$ ◯ B. $\frac{1}{2}(\underline{a}\times\underline{b})$ ◯

C. $|\underline{a}\times\underline{b}|$ ◯ D. $\frac{1}{2}|\underline{a}\times\underline{b}|$ ◯

Federal Board HSSC-II Examination
Mathematics Model Question Paper
(Curriculum 2000)

Time allowed: 2.35 hours | Total Marks: 80

Note: Sections 'B' and 'C' comprise pages 1-2 and questions therein are to be answered on the separately provided Answer Book. Write your answers neatly and legibly.

SECTION – B (Marks 48)

Q.2 Attempt any **TWELVE** parts. All parts carry equal marks. (12 × 4 = 48)

i. If $u(x) = \sqrt{x^2 - 3}$ and $v(x) = x^2 + 3$, then find
a. $u(v(x))$ b. $v(v(x))$

ii. Let $f(x) = \begin{cases} \frac{sin5x}{x} & if\ x < 0 \\ (1-x)^{-\frac{5}{x}} & if\ x \geq 0 \end{cases}$,
Find (a) $\lim_{x\to 0^-} f(x)$ (b) $\lim_{x\to 0^+} f(x)$

iii. If $y = u(x)v(x)$, then find $\frac{dy}{dx}$ by definition.

iv. Differentiate $y = \frac{1+cscx}{1-cscx}$ w.r.t. x.

v. If $x^2 = \frac{x+y}{x-y}$, then find $\frac{d^2y}{dx^2}$.

vi. Evaluate $\int \frac{x}{\sqrt{x-1}} dx$, using substitution method.

vii. Solve $(1 + x^2)y_1 = 1$ with $y(1) = 0$.

viii. Evaluate $\int_{-1}^{1} (1 - |x|)dx$.

ix. The coordinates of the end-points of a line segment PQ are $P(3,7)$ and $Q(11, -6)$. Find the coordinates of point R on the y-axis such that $PR = QR$.

x. For what values of k, the lines $2x - 5 = ky$ and $(k + 1)x = 6y - 3$ have same gradient?

xi. The line joining the points $A(0,6)$ and $B(8,0)$ cuts the line $y = x + 1$ at the point M. Find the coordinates of M.

xii. Find the equation of hyperbola centered at the origin that has a focus at $(3,0)$ and the line $x = 1$ as the corresponding directrix.

xiii. Locate the vertices of an ellipse of eccentricity 0.8 whose foci lie at the points $(0, \pm 8)$. Also find the equation of an ellipse.

xiv. Express $\underline{u} = 3\hat{\imath} - 2\hat{\jmath} + 6\hat{k}$ as a product of its length and direction.

xv. Find the work done by the force $F = \hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ in moving an object along the line from the origin to the point $(2, -1, 4)$.

xvi. Find the area of the triangle whose vertices are $A(1, -1, 0)$, $B(2, 1, -1)$ and $C(-1, 1, 2)$.

SECTION – C (Marks 32)

Note: Attempt any **FOUR** questions. All questions carry equal marks. (4 × 8 = 32)

Q.3 For what value of a and b, the function

$$f(x) = \begin{cases} x^2 - 1 & if\ 2 < x < 3 \\ 2ax & if\ x = 3 \\ bx - 4 & if\ x > 3 \end{cases} \quad \text{is continuous } \forall x.$$

Q.4 If $y = ae^x + be^{2x} + ce^{3x}$, where $a, b, c \in R$, then show that $y_3 - 6y_2 + 11y_1 - 6y = 0$.

Q.5 Evaluate $\int_1^3 \frac{lnx \sin(lnx)}{x} dx$.

Q.6 $A(2,3)$ and $B(3,5)$ are the fixed points and a point P moves such that $\Delta PAB = 8$ square units and P is also equidistant from A and B. Find

a. the equation of the locus of point P.
b. the coordinates of the point P.

Q.7 Mr. Ahmed wishes to mix two types of food in such a way that vitamin contents of the mixture contain at-most 9 units of vitamin A and 12 units of vitamin C. Food A contains 3 units/kg of vitamin A and 2 units/kg of vitamin C. Food B contains 2 units/kg of vitamin A and 3 units/kg of vitamin C. It costs Rs. 50 per kg to buy Food A and Rs. 70 per kg to buy Food B.

a. Write down the three inequalities.
b. Draw graphs on the same axis to show these conditions.
c. Shade the area containing the solution set.
d. Determine the minimum cost.

Q.8 Find the coordinates of the midpoint of the chord cut off from the line $x - 2y + 7 = 0$ by the circle $x^2 + y^2 - 2x - 10y + 1 = 0$.

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MATHEMATICS HSSC-II (3rd Set)

Student Learning Outcomes Alignment Chart

(National Curriculum 2000)

S#	Section: Q. No. (Part no.)	Contents and Scope	Student Learning Outcomes
1	A: 1(i)	Composition and Inversion of Functions	To know the meaning of the identity and constant functions and the techniques of composing and inverting the functions by algebraic methods.
2	A:1(ii)	Limits of important functions	a) limit of functions at x = a $\frac{(x^n - a^n)}{(x - a)}$
3	A: 1(iii)	Limits of important functions	c) limit of functions at x = a $\frac{sinx}{x}$
4	A: 1(iv)	Extreme Values	To find whether a function is increasing or decreasing at a point and in an interval
5	A: 1(v)	Differentiation of Functions other than algebraic	To find derivatives of trigonometric, inverse trigonometric, exponential, logarithmic, hyperbolic and inverse hyperbolic functions.
6	A: 1(vi)	Chain Rule	Explanation and application of chain rule for composite function defined by parametric equations
7	A: 1(vii)	Definite Integrals	To know and apply the theorems of definite integrals
8	A: 1(viii)	Integration by Parts	To know and be able to find the antiderivatives of functions by parts
9	A: 1(ix)	Introduction to Integration	To define integration and to know simple standard integrals.
10	A: 1(x)	Equations of straight lines	g) To know position of a point with respect to a line and to find the distance of a point from a line.
11	A: 1(xi)	Equations of straight lines	g) To know position of a point with respect to a line and to find the distance between two parallel lines
12	A: 1(xii)	Equations of straight lines	a) Concept of the slope of a line. b) To find the slope of a line passing through two points; the equations of the x-axis and y-axis.
13	A: 1(xiii)	Linear Programming	To have the concepts of simple linear programming and of optional solution of the linear objective functions.
14	A: 1(xiv)	Linear Inequalities and their Graphs	To know the meaning of linear inequalities in two variables and their solution be graphically illustrated.
15	A: 1(xv)	Circle	c) to know the general form of equation of circle as $x^2 + y^2 + 2gx + 2fy + c = 0$

			and be able to find its center and radius.
16	A: 1(xvi)	Ellipse and its Elements	To know the concept of an ellipse and its elements (center, foci, eccentricity, vertices, major and minor axes, focal chords, latera racta and directrices)
17	A: 1(xvii)	Parabola and its Elements	To derive the standard forms of equations of parabola.
18	A: 1(xviii)	Introduction of Vector in Space	b) to know direction angles, direction cosines and direction ratios of a vector.
19	A: 1(xix)	Scalar Triple Product of Vectors	d) To find the volume of a parallelopiped and regular tetrahedron.
20	A: 1(xx)	Vector Product of two Vectors	d) To know properties of vector product of two vectors; 3) $1/2\ \lvert a \times b \rvert =$ area of a triangle where a and b represent two sides.
21	B: 2(i)	Composition and Inversion of Functions	To know the meaning of the identity and constant functions and the techniques of composing and inverting the functions by algebraic methods.
22	B: 2(ii)	Limits of important functions	c) limits of the function at x = 0; $(1+x)^x$; $\frac{sinx}{x}$
23	B: 2(iii)	Theorems of Differentiation	To establish the theorems on differentiation sum, difference, product and quotient of functions and their applications
24	B: 2(iv)	Differentiation of Functions other than Algebraic	To find derivatives of trigonometric, inverse trigonometric functions using chain and other rules.
25	B: 2(v)	Successive Differentiation	To find 2nd derivative of implicit functions.
26	B: 2(vi)	Integration by Substitution	To know and be able to integrate by applying the method of substitution in the integration of functions.
27	B: 2(vii)	Differential Equations	To be able to solve differential equations of first order with variables separable in the forms; $dy/dx = f(x)/g(y)$ or $dy/dx = g(y)/f(x)$ concept of initial conditions and simple applications
28	B: 2(viii)	Definite Integrals	c) $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
29	B: 2(ix)	Coordinate System	Be able to; a) locate a point in a Cartesian Plane, b) derive the distance formula
30	B: 2(x)	Equations of straight lines	a) Concept of the slope of a line.
31	B: 2(xi)	Two and three Straight lines	Be able to find; a) the point of intersection of two straight

			lines.
32	B: 2(xii)	Equation of Hyperbola with given elements	To find the equation of hyperbola with focus, center and directrix.
33	B: 2(xiii)	Equation of an Ellipse with given elements	To find the equation of hyperbola with foci, eccentricity and vertices.
34	B: 2(xiv)	Introduction of Vector in Space	a) Fundamental unit vectors (i.j.k) components of a vector$\bar{a} = a_1\underline{i} + a_2\underline{j} + a_3\underline{k}$, magnitude of a vector, unit vector
35	B: 2(xv)	Scalar Triple Product of Vectors	e) Applications of vectors in solving simple problems of mechanics
36	B: 2(xvi)	Vector Product of Two Vectors	3) $1/2\ \lvert a \times b \rvert =$ area of a triangle where a and b represent two sides.
37	C: 3	Continuous and Discontinuous Functions	To understand the concept of continuity of function at a point and in an interval intuitively
38	C: 4	Successive Differentiation	To find 2nd, 3rd, and 4th derivatives of an algebraic, trigonometric, exponential and logarithmic functions
39	C: 5	Definite Integrals (Integration by Substitution)	To be able to differentiate between definite and indefinite integrals and to know and apply; a) $\int_a^b f(x)\,dx$ as the area under the curve. $y = f(x)$ from $x = a$ to $x = b$
40	C: 6	• Coordinate System • Equations of Straight Lines	b) Distance Formula h) To find the area of a triangle whose vertices are given
41	C: 7	Linear Programming	To have the concepts of linear programming and of option solution of the linear objective functions and to find the optimal solution of the linear objective functions by graphical methods
42	C: 8	Tangents and Normals	To find; a) the point of intersection of a circle with a line

MATHEMATICS HSSC-II (3rd Set)

Table of Specification

Topics	1. Functions and Limits	2. Differentiations	3. Integration	4. Introduction to Analytic Geometry	5. Linear Inequalities and Linear Programming	6. Conic Sections	7. Vectors	Total marks for each assessment objective	% age
Knowledge based	1-(1)(1) 2-i(4)	1-(5)(1) 2-iv(4)	1-(8)(1) 1-(9)(1)	1-(10)(1) 1-(11)(1) 2-x(4) 6-(8)	1-(14)(1)	1-(15)(1) 1-(16)(1) 8-(8)	1-(18)(1) 2-xiv(4)	42	31.8%
Understanding based	1-(2)(1), 1-(3)(1) 2-ii(4) 3-(8)	2-iii(4) 2-v(4) 4-(8)	2-vi(4)	2-xi(4)	1-(13)(1) 7-(8)	1-(17)(1) 2-xii(4) 2-xiii(4)	1-(19)(1) 1-(20)(1) 2-xvi(4)	62	47%
Application based		1-(4)(1) 1-(6)(1)	1-(7)(1) 2-vii(4) 2-viii(4) 5-(8)	1-(12)(1) 2-ix(4)			2-xv(4)	28	21.2%
Total marks for each topic	19	23	23	23	10	19	15	132	100%

KEY:

1(1)(01)

Question No (Part No.) (Allocated Marks)

Note: (i) The policy of FBISE for knowledge based questions, understanding based questions and application based questions is approximately as follows:

a) 30% knowledge based.
b) 50% understanding based.
c) 20% application based.

(ii) The total marks specified for each unit/content in the table of specification is only related to this model question paper.

(iii) The level of difficulty of the paper is approximately as follows:

a) 40% easy
b) 40% moderate
c) 20% difficult